

AP–NMS Integration Interface Specification

1. Purpose

This document describes the minimum interface required for integrating an Access Point (AP) system with an external Network Management System (NMS). The goal of this integration is to allow the NMS to: 1. Receive periodic status reports from each AP. 2. Request traffic statistics from each AP. 3. Send control commands to APs. 4. Receive command execution acknowledgements. This interface does not replace or modify the original AP system. Instead, it provides a lightweight integration layer that communicates with the NMS. The AP system continues to run its own internal software as usual.

2. Network Requirements

The NMS runs an HTTP service on port 8000. The integration code running on the AP automatically searches for the NMS service on the local network. Therefore: - Each AP must be on the same LAN as the NMS. - The AP must be able to reach the NMS service via HTTP on port 8000. No inbound connections to the AP are required. All communication is initiated by the AP.

3. Registration and AP Identity

Each AP must register with the NMS when it starts. During registration: 1. The AP sends its MAC address to the NMS. 2. The NMS checks a whitelist to determine whether the AP is allowed to join. 3. If allowed, the NMS returns an assigned AP name.

Example assigned name:
ap08

Hostname Enforcement: If the AP hostname does not match the assigned name, the integration layer will: 1. Change the system hostname to the assigned name 2. Reboot the AP This typically happens only once when the AP is first integrated into the system. If renaming the system hostname is not acceptable for the AP platform, the AP can alternatively keep its existing hostname and simply report that name to the NMS.

4. Overall Architecture

The AP–NMS interaction uses a client-pull model. Each AP periodically contacts the NMS to: - report status - retrieve commands - send execution results Advantages: - The AP does not expose any inbound API. - The NMS only runs a single HTTP service. - The system works well in restricted network environments.

5.1 Control + Status Channel

Interval: every 10 seconds

Endpoint:

POST /ap/poll/{ap_id}

Example request body:

{

```

"time": "2026-03-12-10:30:00",
"status": {
  "mac": "aa:bb:cc:dd:ee:ff",
  "ip": "192.168.1.20",
  "ssids": ["Office-5G"],
  "band": "5g",
  "channel": 44,
  "antenna_count": 2,
  "associations": [
    {
      "sta_mac": "11:22:33:44:55:66",
      "ssid": "Office-5G",
      "mcs": 7
    }
  ]
}
}

```

5.2 Traffic Statistics Channel

Interval: once per minute when enabled.

Endpoint:

POST /ap/traffic/{ap_id}

Example request body:

```

{
  "time_start": "2026-03-12-10:29:00",
  "time_end": "2026-03-12-10:30:00",
  "records": [
    {
      "sta_mac": "11:22:33:44:55:66",
      "ac": "BE",
      "avg_frame_duration_us": 820.5,
      "frame_count": 1200,
      "mcs_distribution": {
        "0": 15,
        "1": 30,
        "2": 100,
        "3": 180,
        "4": 250,
        "5": 300,
        "6": 220,
        "7": 105
      }
    }
  ]
}

```

Statistics are reported per STA and per QoS Access Category (AC). Aggregate AP statistics are not required.

6. Command Execution

Commands are delivered through the poll response.

Example command:

```
{
  "category": "ap",
  "action": "ap.txpower.set",
  "args_json": "{\"txpower\":10}"
}
```

Command acknowledgement endpoint:

POST /cmd/ack/{ap_id}

Example acknowledgement:

```
{
  "cmd_id": "1773133049125-0",
  "status": "ok",
  "detail": "tx power updated"
}
```

7. Initial Command Set

Association management:

```
ap.association.get
ap.sta.associate
ap.sta.disassociate
```

Transmit power control:

```
ap.txpower.set
```

Traffic reporting control:

```
ap.traffic.enable
ap.traffic.disable
```

8. AP Integration Layer

The integration layer performs the following tasks: 1. Register with the NMS when the AP starts. 2. Periodically send status reports. 3. Poll the NMS for commands. 4. Execute commands using the AP system. 5. Send acknowledgements. 6. Send periodic traffic reports. This layer does not modify the AP system. It only acts as a communication bridge between the AP and the NMS.

9. Interfaces Required from the AP System

The AP system must provide: Status Provider: - MAC address - IP address - active SSIDs - band - channel - antenna count - association table Traffic Statistics Provider: - average frame duration - total frame count - MCS distribution Command Execution Interface: - STA association - STA disassociation - transmit power adjustment

10. Deployment Model

The integration layer is delivered as a software bundle. Installation performs: 1. Copy integration files to the system 2. Configure runtime environment 3. Provide entry points for starting the NMS interaction The AP platform may decide when the integration process starts and how logs are handled. The integration layer does not automatically start itself after installation.

11. Extensibility

The architecture allows new commands and features to be added later without redesigning the system. Examples:
- channel selection - bandwidth configuration - radio enable/disable - advanced statistics

12. Summary

The AP–NMS integration layer provides a communication bridge between AP devices and the NMS. Key characteristics: - AP initiates all communication. - Status is reported every 10 seconds. - Traffic statistics are reported once per minute when enabled. - Commands are delivered through polling. The AP system only needs to provide status information, traffic statistics, and command execution capability.